	LABORATORY EXERCISE **ELECTRICAL ENGINEERING STUDIES** **COLLEGE OF ENGINEERING** **UNIVERSITI TEKNOLOGI MARA** **CAWANGAN PULAU PINANG**	

COURSE NAME:	ELECTRONICS CIRCUIT ANALYSIS	COURSE CODE:	ETL512
PROGRAMME CODE:	CEEE211	CO – PO	CO3 – PO5
EXPERIMENT:	SINGLE PHASE RECTIFIER	EXPERIMENT DATE:	
LECTURER'S NAME:	MADAM ROSHEILA DARUS	SUBMISSION DATE:	2/3/2026

STUDENT'S NAME	GROUP	STUDENT ID
MUHAMMAD DANIEL HAKIM BIN ZAMZUHARI	PEE2113A2	2024407158

LECTURER'S COMMENT (if any):

CLO3		Construct various electronic circuit analysis using modern engineering software. (P4)						
PO5 (ETAC)		Select and apply/perform while recognising limitations of appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to broadly-defined electronics engineering technology problems. (SK2 and SK6)						
CRITERIA	**SUB-ATTRIBUTE**	**WEAK**	**POOR**	**FAIR**	**GOOD**	**SCORE**	**WEIGHT**	**MARKS**
		0	1 – 2	3 – 4	5			
	Installation Software							

Experimental/ Demonstration setup/ Simulation	Able to follow instructions and install PSIM Software from the website.	Unable to install the software from the website. .	Able to install the software with considerable amount of assistance.	Able to install the software with very little assistance.	Able to install the software without assistance and the software can be launched properly.		0.5	
	<u>Conducting Experiment 5a i</u> **[SP1 SK3 SK4]**							
	Able to construct circuit using PSIM	Unable to construct or operate the circuit even with assistance. No evidence of software	Attempted to build the circuit, but major errors persist. Software usage is	Circuit partially functional. Some components connected correctly but lacks clarity or optimization. Software usage	Circuit fully optimized, neat, and error-free. Tool usage is precise and efficient. Shows mastery of circuit design		1	

		and circuit understanding	unclear or incorrect.	shows limited understanding	and software tool.			

	Able to use the software correctly, and generate the required output by utilizing the appropriate tools available in the software	Unable to use tools in the software to generate the output	Tried to use the tools available in the software but selected the wrong tool	Able to use the tools available in the software, but applied them incorrectly to obtain the result	Able to use the correct tools and obtain the correct output result		1	
	<u>Conducting Experiment 5a ii</u> **[SP1 SK2 SK4] [SP4]**							
	Able to construct circuit using PSIM	Unable to construct or operate the circuit even with assistance. No evidence of software and circuit understanding	Attempted to build the circuit, but major errors persist. Software usage is unclear or incorrect.	Circuit partially functional. Some components connected correctly but lacks clarity or optimization. Software usage shows limited understanding	Circuit fully optimized, neat, and error-free. Tool usage is precise and efficient. Shows mastery of circuit design and software tool.		1	
	Able to use the software correctly, and generate the required output by utilizing the appropriate tools available in the software	Unable to use tools in the software to generate the output	Able to use the tools available in the software but selected the wrong tool	Able to use the tools available in the software, but applied them incorrectly to obtain the result	Able to use the correct tools and obtain the correct output result		1	

	<u>Conducting Experiment 5b i</u> **[SP1 SK3 SK4] [SP4]**							
	Familiarity issue using same circuit topology for different types of semiconductors	Unable to construct or operate the circuit even with assistance. No evidence of software and circuit understanding	Attempted to build the circuit, but major errors persist. Software usage is unclear or incorrect.	Circuit partially functional. Some components connected correctly but lacks clarity or optimization. Software usage shows limited understanding	Circuit fully optimized, neat, and error-free. Tool usage is precise and efficient. Shows mastery of circuit design and software tool.		1	
	Able to use the software correctly, and generate the required output by utilizing the appropriate tools available in the software	Unable to use tools in the software to generate the output	Able to use the tools available in the software but selected the wrong tool	Able to use the tools available in the software, but applied them incorrectly to obtain the result	Able to use the correct tools and obtain the correct output result		1	
	<u>Conducting Experiment 5b ii</u> **[SP1 SK3 SK4] [SP4]**							
	Common conflicting requirements to ensure that the	Unable to construct or operate the circuit even	Attempted to build the circuit, but major errors	Circuit partially functional. Some components connected	Circuit fully optimized, neat, and error-free. Tool usage is		1	

	output remains within the specified condition	with assistance. No evidence of software and circuit understanding	persist. Software usage is unclear or incorrect.	correctly but lacks clarity or optimization. Software usage shows limited understanding	precise and efficient. Shows mastery of circuit design and software tool.			
	Able to use the software correctly, and generate the required output by utilizing the appropriate tools available in the software	Unable to use tools in the software to generate the output	Able to use the tools available in the software but selected the wrong tool	Able to use the tools available in the software, but applied them incorrectly to obtain the result	Able to use the correct tools and obtain the correct output result		1	
Data Collection and analysis Experiment 5a & 5b **[SP1 SK3 SK4][SP2][SP4]**	**<u>Data Collection</u>** Able to measure data completely and achieve experiment objectives	Unable to record any result	Incomplete data measurement with correct /incorrect value	Complete data measurement but some are incorrect	All data are measured and correct		0.5	
	<u>Data analysis</u> Interpretation of results and handling of invalid inputs	No analysis or incorrect conclusions	Basic analysis; not linked to theory	Analysis present; partially linked to theory	Insightful analysis; includes invalid input handling and theoretical grounding		0.5	

	<u>Data Evaluation & Improvement</u>	No problemsolving attempt	Identifies issue but no resolution	Suggests basic improvement	Clearly identifies challenges (e.g., invalid		0.5	
	Ability to address conflicting requirements and propose improvements				inputs) and proposes viable solutions			
								/50



ELECTRICAL ENGINEERING STUDIES

COLLEGE OF ENGINEERING

UNIVERSITI TEKNOLOGI MARA CAWANGAN PULAU PINANG

ELECTRONICS CIRCUIT ANALYSIS (ETL512)

EXPERIMENT 5
SINGLE PHASE RECTIFIER
(SOFTWARE SIMULATION-OPEN ENDED)

OBJECTIVES

1. To learn how to use PSIM as an engineering tool.
2. To examine the difference between diode rectifier and silicon-controlled rectifier.
3. To verify the output voltage between the simulation and theory.

LIST OF APPARATUS

1. PSIM software

THEORY

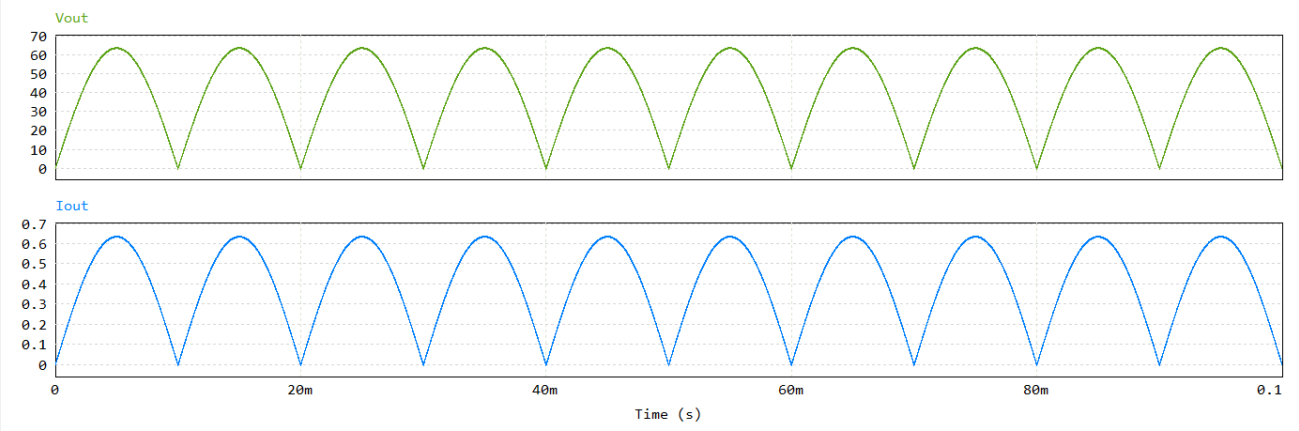
A conventional **full-wave diode rectifier cannot actively control its output voltage because Diodes conduct only when forward-biased**. There is no mechanism to **adjust conduction time** (diodes switch on/off based solely on input polarity). On the other hand, to achieve **adjustable DC output**, replace diodes with **active switching devices such as** SCRs **(Silicon-Controlled Rectifier)**. An **SCR** controls the average output voltage by **delaying the conduction angle.**

Report Experiment 5 a)

i) A single phase full-wave diode rectifier produces with input voltage 45 V_{rms}. The rectifier is connected to the 100 Ω load. Simulate this circuit using PSIM software. [SP1 SK3 SK4]

Question [SP1 SK3 SK4]

1. Plot and measure the output voltage and output current of the rectifier.



2. Compare your findings with the experimental result in experiment 3. Comment on your finding.

Efficiency: Full-Wave (Simulation) is more efficient because it uses both cycles (positive and negative). Half-Wave (Exp 3) only uses half the cycle, so a lot of energy is "wasted".

3. Verify your output voltage and current from the simulation and experimental result with theoretical calculation. Comment on your finding.

Voltage output

$$Vdc = \frac{2Vpeak}{\pi}$$

$$= \frac{2(45 * \sqrt{2})}{\pi} = 40.51V$$

Current output

$$Idc = \frac{Vdc}{R} = \frac{2Vpeak}{\pi * R}$$
$$= 0.405A$$

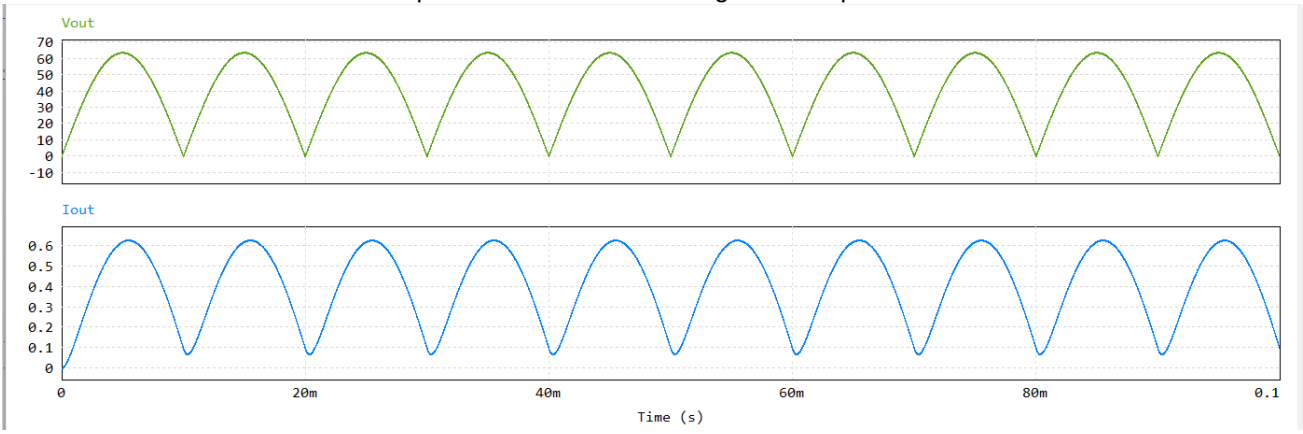
In the previous experiment 3, we used a Half-Wave Thyristor, the voltage only out half and we can control the current flow using firing angle inside thyristor.

In Full-Wave Diode, voltage came out full (positive and negative). The difference is that circuit cannot be controlled because the diode will send current instantly when voltage input.

ii) A single phase full-wave diode rectifier produces with input voltage 45 V_{rms}. The rectifier is connected to the 100 Ω load and 50 mH [SP1 SK3 SK4][SP 4]

Question [SP1 SK3 SK4][SP4]

1. Plot and measure the output waveform of the voltage and output current of the rectifier.



2. Compare your findings with the experimental result in experiment 3. Comment on your finding.
   This simulation output is (Full-Wave) gives two voltage cycles, while Exp 3 only gives one.
   In this Full-Wave simulation, current becomes more continuous. In Exp 3 (Half-Wave), the inductor cause the current to flow longer, but the effect is not significant.

3. Verify your output voltage and current from the simulation and experimental result with theoretical calculation. Comment on your finding.

Voltage output

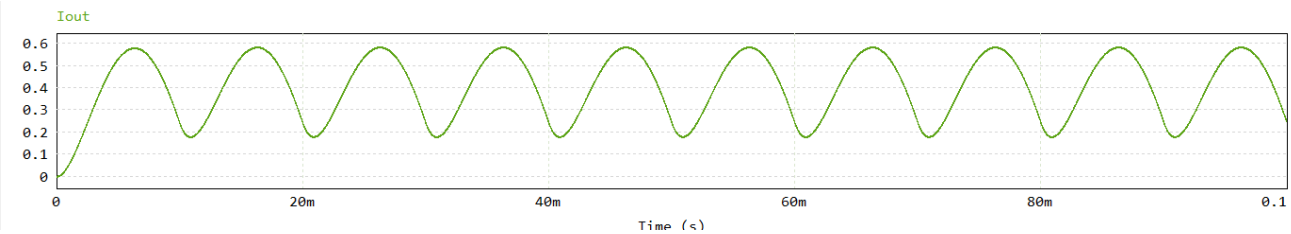
$$Vdc = \frac{2Vpeak}{\pi}$$

$$= \frac{2/(45 * \sqrt{2})}{\pi} = 40.51V$$

Current output

$$Idc = \frac{Vdc}{R} = \frac{2Vpeak}{\pi * R}$$
$$= 0.405A$$

4. What will happen to the output if the inductor of the load increases to 150 mH. Comment on your finding.



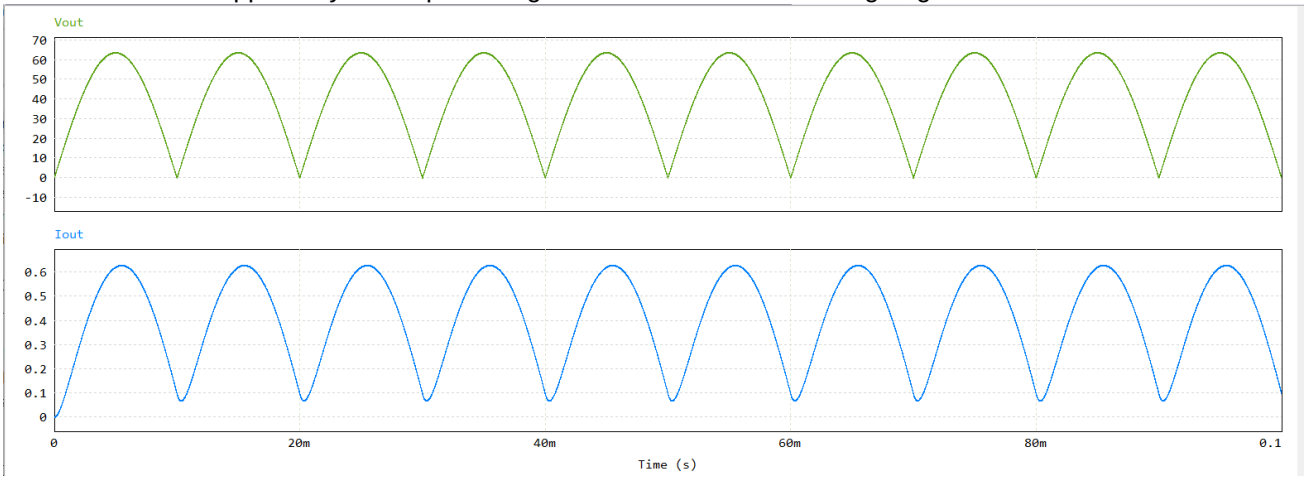
When the inductor is increased to 150 mH, the current ripple becomes smaller and the output current becomes more smooth/stable. This happens because the larger inductor has higher inductance properties to resist current changes. However, the average values of voltage and current remain unchanged because they only depend on the input value and load resistance.

Experiment 5 b)

i) Using the same circuit as in Experiment 5a, replace the diode with an SCR. Insert a pulse generator/trigger circuit to control the SCR firing angle. [SP1 SK4 SK3], [SP2], [SP 4]

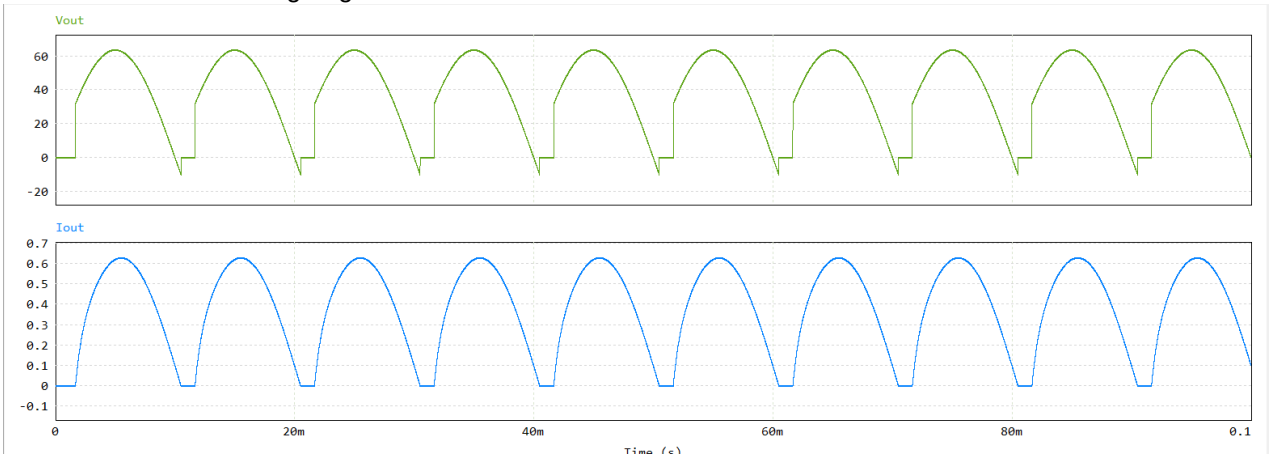
Question

1. What will happen to your output voltage and current when the firing angel is 0°?

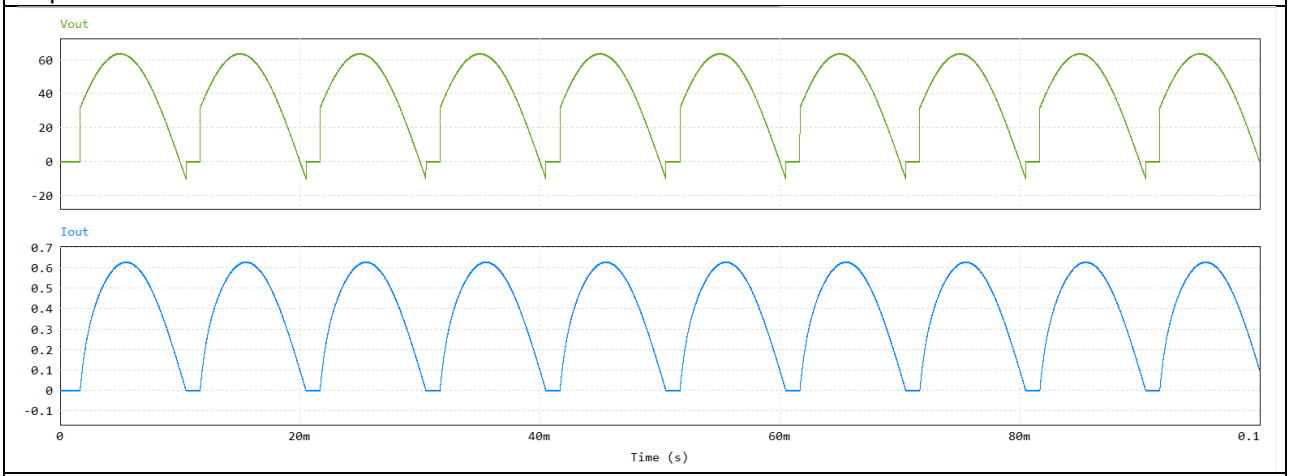


When the firing angle is set to 0°, the SCR will behave like a normal diode. The output voltage and current will be at maximum and the waveform will be full (no clipping).

2. Increase the firing angle to 30°.

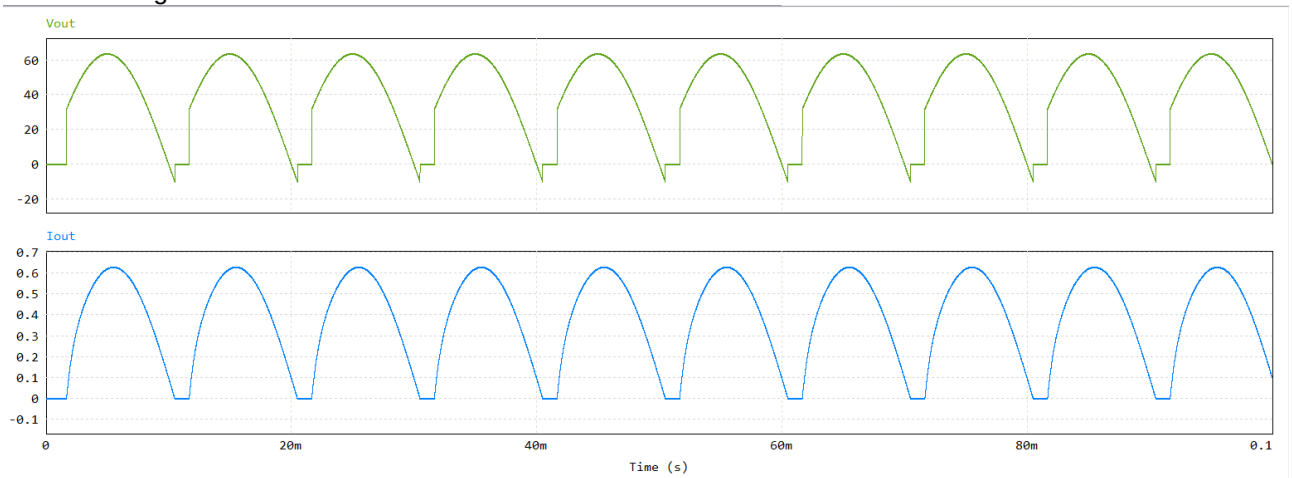


3. Compare your findings with the experimental result in experiment 4. Comment on your finding.

Experiment 5b

Experiment 4

The voltage and current waveforms in the PSIM simulation are in almost same with the results of Experiment 4. Both show a 'clipped' effect at the beginning of the cycle due to the 30° firing angle.

4. Plot and measure the output waveform of the voltage and current of the rectifier. Comment on your finding.



This is the effect of the 50mH inductor which acts to store energy and smooth out the current flow.

5. Verify your output voltage and current from the simulation and experimental result with theoretical calculation. Comment on your finding.

Voltage output

$$\alpha = \text{Firing angle}$$

$$Vdc = \frac{2Vpeak}{\pi}\cos\alpha$$

$$= \frac{2 * 63.64}{\pi}\cos(30°) = 35.08V$$

Current output

$$Idc = \frac{Vdc}{R}$$

$$= \frac{35.08V}{100\ \Omega} = 0.351A$$

This finding also confirms that the firing angle alpha functions as a power regulator. When alpha is increased from 0 to 30 the output voltage decreases because part of the voltage cycle is 'blocked' by the SCR. This shows that this system is very useful for applications that require adjustable voltage (adjustable DC output)

6. Adjust the firing angle of the SCR to produce 35 V_{avg} output voltage. Comment on your finding.

$$35 = 40.51 \times \cos(\alpha)$$

$$\cos(\alpha) = \frac{35}{40.51}$$

$$\cos(\alpha) = 0.864$$

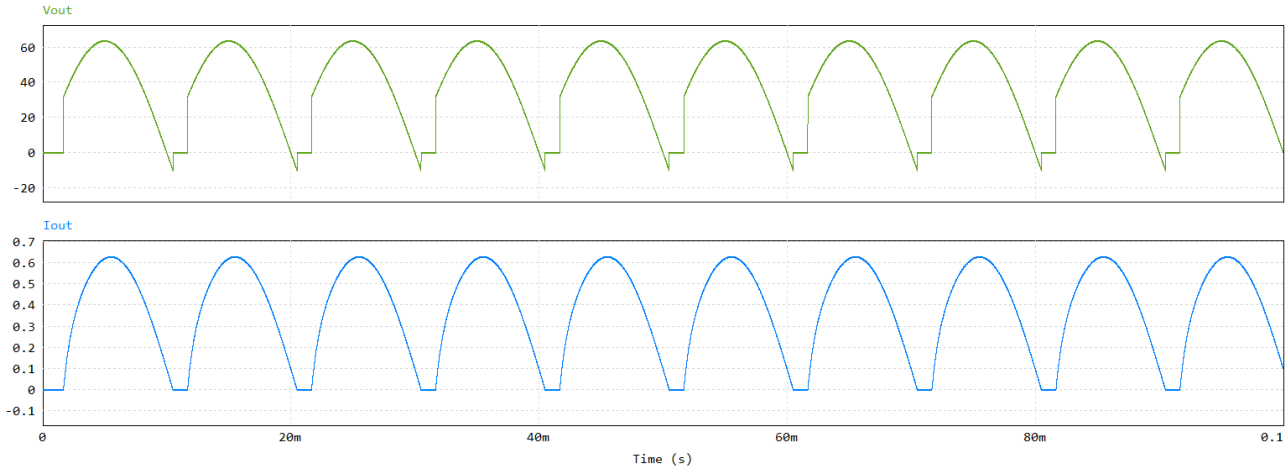
$$\alpha = \cos^{-1}(0.864) \approx \mathbf{30.23}^{\circ}$$

Even though the output voltage has a "blank" part (0V during the delay), the output current (Iout) does not fall to zero. Due to the presence of the Inductor (L) the current is flow continuously with a smoother shape. The inductor acts to provide stored energy when the SCR is in the OFF state.

ii) From experiment 5b i) change the load type. The new parameter for the load is 100 Ω resistor and 50 mH inductor and firing angle in this experiment is 30°. [SP 1 SK3 SK4][SP 2][SP4]

Question

1. Plot and measure the output voltage and output current of the rectifier. Comment on your finding.



The output voltage is pulsating, the output current (Iout) not drop to zero. The 50 mH inductor causes the current flow even the input voltage has dropped. The inductor is to store energy in the form of magnetic field and release it, it's smoothing out current ripple so it's becomes more stable and continuous.

2. Compare your findings with the experimental result in experiment 4. Comment on your finding.

The voltage waveforms in both experiments were similar, with 'delay' or cutoff at the beginning of the cycle. This shows that SCR in both conditions (simulation and laboratory) works to control the voltage phase in same way.

3. Verify your output voltage and current from the simulation and experimental result with theoretical calculation. Comment on your finding.

Voltage output

$$\alpha = \text{Firing angle}$$

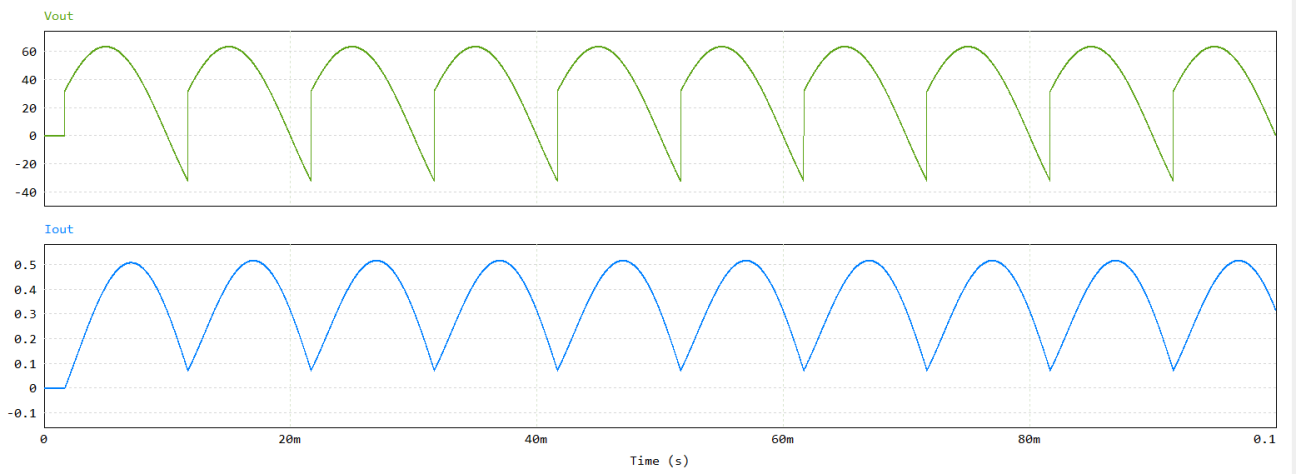
$$Vdc = \frac{2Vpeak}{\pi}\cos\alpha$$
$$= \frac{2 * 63.64}{\pi}\cos(30°) = 35.08V$$

Current output

$$Idc = \frac{Vdc}{R}$$
$$= \frac{35.08V}{100\ \Omega} = 0.351A$$

The use of SCR allows us to control the output voltage by adjusting firing angle alpha. The larger the firing angle, the lower average voltage produced.

4. What will happen if the inductor of the load increases to 250 mH. Comment on your finding.



The current ripple will be more reduced. The current will look almost like a straight line.
When the inductance value is increased to 250 mH, a significant decrease in current ripple. More larger inductors have higher ability to store energy and resist any changes in current flow.

5. What is the minimum value of the load inductor to make sure the output current conduction is in continuous mode.

$$\omega = 2\pi f = 2 * \pi * 50 = 314.16 \frac{\text{rad}}{\text{s}}$$

$$\text{Lmin} = \frac{R * \tan(\alpha)}{\omega}$$

$$= \frac{100 * \tan(30)}{314.16} = 0.1837\text{H} = 183.7\text{mH}$$

50 mH Load = 50mH < 183.7mH, current is actually Discontinuous mode and very close to zero line

250 mH Load = 250 mH > 183.7mHcurrent is in Continuous mode. That current graph looks much smoother than 50 mH.

Results

Interpret your results and suggest improvements or further experiments.

Interpretation : Successfully controls voltage by adjusting the firing angle (alpha).
Inductor: Converts current from discontinuous to continuous when the value of L is increased above 183.7 mH.
We can test a Three-Phase circuit for higher and more stable power in next experiments.
Use Microcontroller like Arduino/ESP32 for more precise digital and remotely control of the trigger angle.

